

Chenning Wu

Basic Information



Name: Wu Chenning

Affiliation: College of Computer Science and Artificial Intelligence, Fudan University
(Expected graduation: June 2027)

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Education

- **2017 – 2021**
Fudan University, School of Software — *B.Sc. in Software Engineering*
GPA: 3.68/4.0 (Top 5%), Excellent Graduate Award. UC Davis | Exchange Program (Quarter)
- **2021 – Present**
Fudan University, School of Computer Science and Technology — *Ph.D. Candidate in Big Data and Data Science*
Research Topic: AI4DB -- Research on Automated Index Selection Techniques for Hybrid Scalar-Vector Scenarios

Research Projects

- **Learned Cost Model for Index Tuning on Approximate Nearest Neighbor Search (*On progress*)**
Leveraging learned vector search models to enable adaptive ef_search during query processing.
- **Adaptive Indexing for Predicate-Agnostic Filtered ANN Search (*On progress*)**
- **RIB: Robust Learning-based Index Benefit Estimation (*SIGMOD 2026, Accepted*)**
Utilizing Bidirectional GNNs and Quantile Regression to improve the robustness of index benefit estimation.
- **UTune: Towards Uncertainty-Aware Online Index Tuning (*ICDE 2026, Accepted*)**
Proposing an uncertainty-aware operator-level learned model for dynamic online index tuning.
- **GeoPhrase Tree: An Efficient Index for Frequent Phrase Query over Spatio-temporal Ranges (*DASFAA 2026, Accepted*)**
Achieving efficient phrase queries over spatio-temporal datasets by integrating Suffix Trees with R-trees.

Industry Experience

- **University–Industry Collaboration with Xiaohongshu (Little Red Book)**
Led a research group in the first Fudan–Xiaohongshu collaboration project. Responsible for intelligent database index recommendation research and patent application.
- **The 2nd PolarDB Vector Database Optimization Competition**
Optimized HNSW index efficiency and improved retrieval performance on PoladDB.